

Hibernate ID Generation Strategies

GenerationType.AUTO, IDENTITY, SEQUENCE, TABLE

1. GenerationType.AUTO

What is it?

Hibernate chooses the ID generation strategy automatically based on the database and Hibernate configuration.

When to use it?

Use it when you do not need to control the exact generation mechanism, or when the application may work with different databases.

Example

```
@Id
@GeneratedValue(strategy = GenerationType.AUTO)
private Long id;
```

Database

Can be used with most relational databases. The actual mechanism depends on the database/dialect.

2. GenerationType.IDENTITY

What is it?

The database generates the ID using an identity column. Hibernate asks the database to generate the value when a new row is inserted.

When to use it?

Use it when your database supports identity columns and you want the database itself to manage the generated IDs.

Example

```
@Id
@GeneratedValue(strategy = GenerationType.IDENTITY)
private Long id;
```

Database

Common with MySQL, SQL Server, PostgreSQL, and modern Oracle versions that support identity columns.

3. GenerationType.SEQUENCE

What is it?

Uses a database sequence to generate unique ID values. A sequence is a database object that produces numbers in order.

When to use it?

Use it when the database provides sequences and you want good control and efficient ID generation.

Example

```
@Id
@GeneratedValue(
    strategy = GenerationType.SEQUENCE,
    generator = "teacher_seq"
)
@SequenceGenerator(
    name = "teacher_seq",
    sequenceName = "teacher_seq",
    allocationSize = 1
)
private Long id;
```

Database

Excellent choice for Oracle and PostgreSQL.

4. GenerationType.TABLE

What is it?

Hibernate uses a database table to keep track of the next ID value instead of relying on an identity column or sequence.

When to use it?

Use it when you need a database-independent generation mechanism or when identity/sequence mechanisms are not suitable. It is less common in modern applications.

Example

```
@Id
@GeneratedValue(strategy = GenerationType.TABLE)
private Long id;
```

Database

Can work with many relational databases because it relies on a table.

Comparison

Strategy	Who generates ID?	Best use	Suitable databases
AUTO	Hibernate chooses	When exact strategy is not important	Most relational DBs
IDENTITY	Database identity column	Simple DB-managed IDs	MySQL, SQL Server, PostgreSQL, Oracle
SEQUENCE	Database sequence	Efficient and controllable generation	Oracle, PostgreSQL
TABLE	Hibernate + generator table	Database-independent compatibility	Most relational DBs

Quick interview summary

AUTO: Hibernate decides the strategy automatically.

IDENTITY: The database generates the ID using an identity column.

SEQUENCE: The database generates IDs using a sequence.

TABLE: Hibernate uses a database table to generate IDs.

For Oracle: SEQUENCE is usually the natural choice because Oracle provides sequence objects and Hibernate can use them directly.